

# Тригонометрические тождества, суммы и произведения

Домашнее задание

1 сентября 2026 года

**Теоретический минимум.** The complex exponential and geometric summation; the Dirichlet kernel and boundedness of partial sums; telescoping through  $\cot t - \cot 2t = \frac{1}{\sin 2t}$ ; Vieta's doubling product; the values  $\cot \frac{k\pi}{n}$  as roots of an explicit polynomial; Abel summation.

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1. (закон Морри) Prove  $\cos 20^\circ \cos 40^\circ \cos 80^\circ = \frac{1}{8}$  and, more generally,  $\prod_{k=0}^{n-1} \cos(2^k \theta) = \frac{\sin(2^n \theta)}{2^n \sin \theta}$ ; describe all  $\theta$  for which the product equals  $2^{-n}$ .
2. Prove  $\sum_{k=1}^n \frac{1}{\sin k\theta \sin(k+1)\theta} = \frac{\cot \theta - \cot(n+1)\theta}{\sin \theta}$  and find the analogous telescoping formula for  $\sum \frac{1}{\cos k\theta \cos(k+1)\theta}$ .
3. (MIPT 2019) Prove the discrete Wirtinger inequality

$$\sum_{k=1}^n (x_k - x_{k-1})^2 \geq 4 \sin^2 \frac{\pi}{2n} \sum_{k=0}^n x_k^2$$

for every real sequence  $x_0, \dots, x_n$  with  $x_0 = x_n = 0$ , and show that the constant cannot be improved. (*Подсказка: собственные значения трёхдиагональной матрицы.*)

4. Prove  $\sum_{k=1}^{n-1} \cot^2 \frac{k\pi}{n} = \frac{(n-1)(n-2)}{3}$  and  $\sum_{k=1}^{n-1} \frac{1}{\sin^2 \frac{k\pi}{n}} = \frac{n^2-1}{3}$ .
5. Prove that  $\sum_{k=0}^{n-1} \cos\left(x + \frac{2\pi k}{n}\right) = 0$  and  $\prod_{k=0}^{n-1} \sin\left(x + \frac{k\pi}{n}\right) = 2^{1-n} \sin nx$ .
6. (Proposed problems) Find all polynomials  $P \in \mathbb{R}[x]$  such that  $P(\sin x) + P(\cos x) = 1$  for all real  $x$ .

7. (Mexmat, 2008) Prove that for increasing functions  $f, g: [0, \frac{\pi}{2}] \rightarrow \mathbb{R}$

$$\int_0^{\pi/2} f(x)g(x) \sin x \, dx \geq \int_0^{\pi/2} f(x) \sin x \, dx \int_0^{\pi/2} g(x) \sin x \, dx.$$

8. Prove that  $\tan A + \tan B + \tan C = \tan A \tan B \tan C$  whenever  $A + B + C = \pi$  and none of the tangents is infinite; deduce the identity  $\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$  and use it to prove  $\tan A \tan B \tan C \geq 3\sqrt{3}$  for an acute triangle.
9. (MIPT 2021) Prove that the function  $f(x) = \cos(x - x_1) \cos(x - x_2) \cdots \cos(x - x_n)$  attains its maximal modulus at a point which is at distance at least  $2^{-n}$  from each of the sets  $\{x_i + \frac{\pi}{2} + \pi k : k \in \mathbb{Z}\}$  of its zeros.
10. Prove that  $\sum_{k=1}^n \frac{\sin kx}{k} > 0$  for all  $x \in (0, \pi)$  and all  $n$  (неравенство Фейера–Йенсена). (Подсказка: индукция по  $n$  и оценка через ядро Дирихле.)